Four-Vectors and Spacetime Diagrams

PHYS 2203 — Introductory Physics III

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Overview

Writing the Lorentz transformation in terms of ct rather than t puts space and time on the same footing and leads to the **four-vector**, whose *invariant* — the **spacetime interval** — every observer agrees on. With ct on the vertical axis we can then *draw* the transformation. On the resulting **spacetime diagram** every object traces a *worldline* and light travels at 45°. Those 45° lines form the **light cone**, which sorts the events around any given event into those whose time order every observer agrees on and those whose order is frame-dependent — the causal structure of spacetime.

1 Recap: The Lorentz Transformations

Frame O' moves in the $+x$ direction at constant speed v relative to frame O, and the origins coincide at $t = t' = 0$. Last lecture we derived, from the two postulates alone, the rule relating the coordinates of a single event in the two frames:

$$\boxed{\begin{aligned} x' &= \gamma\,(x - vt), \\ y' &= y, \\ z' &= z, \\ t' &= \gamma\Big(t - \frac{v}{c^2}\,x\Big), \end{aligned}} \qquad \text{where} \qquad \gamma \equiv \frac{1}{\sqrt{1 - v^2/c^2}}. \tag{1}$$

The inverse follows by exchanging primed and unprimed coordinates and reversing the sign of v, since O recedes from O' at $-v$:

$$x = \gamma\,(x' + vt'), \qquad y = y', \qquad z = z', \qquad t = \gamma\Big(t' + \frac{v}{c^2}\,x'\Big). \tag{2}$$

2 The Fourth Dimension: Four-Vectors

The Lorentz transformation mixes space and time together. This section makes that mixing explicit and shows that time is best regarded as a *fourth dimension* alongside x, y, and z.

A symmetric form of the transformation

The transformation looks lopsided as written, because x' carries a vt while t' carries a vx/c^2. The asymmetry is an illusion of units. Measure time in units of length by using ct in place of t:

$$\boxed{\begin{aligned} x' &= \gamma\Big(x - \frac{v}{c}\,ct\Big)\,, \\ y' &= y, \\ z' &= z, \\ ct' &= \gamma\Big(ct - \frac{v}{c}\,x\Big)\,. \end{aligned}} \tag{3}$$

The first and fourth equations are now identical under the exchange of x and ct. Since time mixes with position in exactly the way position mixes with time, **time must be treated as a fourth dimension** on the same footing as the three spatial ones.

Matrix form

Written as a matrix acting on a column of four coordinates,

$$\begin{bmatrix} x' \\ y' \\ z' \\ ct' \end{bmatrix} = \begin{bmatrix} \gamma & 0 & 0 & -\gamma\,\dfrac{v}{c} \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -\gamma\,\dfrac{v}{c} & 0 & 0 & \gamma \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ ct \end{bmatrix}. \tag{4}$$

The matrix is symmetric, and its structure makes the four-dimensional character of spacetime apparent. It is often the most convenient way to apply all four equations at once.

Definition of a four-vector

Four-vector. Any four quantities (A_x, A_y, A_z, A_t) that transform from one inertial frame to another by the Lorentz transformation constitute a *four-vector*.

We already have one example: the **position-time four-vector** (x, y, z, ct). For a general four-vector the transformation reads

$$A'_x = \gamma\Big(A_x - \frac{v}{c}A_t\Big)\,, \qquad A'_y = A_y, \qquad A'_z = A_z, \qquad A'_t = \gamma\Big(A_t - \frac{v}{c}A_x\Big)\,. \tag{5}$$

Other four-vectors appear later in the course. In particular, energy and momentum combine into a four-vector when we discuss relativistic dynamics in Lecture 17.

The invariant

Every four-vector carries with it a quantity that *all* observers agree on. From the transformation above one can show that

$$\boxed{A'^2_t - \big(A'^2_x + A'^2_y + A'^2_z\big) = A_t^2 - \big(A_x^2 + A_y^2 + A_z^2\big)\,.} \tag{6}$$

The time component squared minus the sum of the squares of the spatial components has the same value in every inertial frame. Such a quantity is called an **invariant**. Writing $\mathbf{A}$ for the spatial part, this is $A'^2_t - \mathbf{A}'\cdot\mathbf{A}' = A_t^2 - \mathbf{A}\cdot\mathbf{A}$.

The invariant of the position-time four-vector. A *difference* between two events obeys the same transformation, so apply the result to $(\Delta x, \Delta y, \Delta z, c\,\Delta t)$:

$$\boxed{(\Delta s)^2 \equiv (c\,\Delta t)^2 - \left(\Delta x^2 + \Delta y^2 + \Delta z^2\right) = \text{the same in every frame.}} \tag{7}$$

This combination is the **spacetime interval** between the two events. Although observers disagree about the time separation Δt and about the spatial separation, they all agree on this particular combination of the two.

What the invariant means. Evaluate it in the most convenient frame — the one in which the two events occur at the *same place*, so that $\Delta x = \Delta y = \Delta z = 0$. There the interval is simply $(c\,\Delta t_0)^2$, where Δt_0 is the proper time. Hence

$$(c\,\Delta t)^2 - \left(\Delta x^2 + \Delta y^2 + \Delta z^2\right) = (c\,\Delta t_0)^2. \tag{8}$$

Suppose the two events are successive heartbeats of a space traveler. In her own frame they happen at the same place and are separated by her proper time. An observer watching her fly past assigns them different positions and a longer time separation, exactly as time dilation requires — but both observers compute the same interval, and so agree on how many times her heart has beaten. The proper time between a pair of events does not depend on who is watching.

3 Spacetime Diagrams and Worldlines

A **spacetime diagram** (or *Minkowski diagram*) plots position x horizontally and time vertically. We plot ct rather than t on the vertical axis, so that both axes carry units of length.

Worldline. The *worldline* of an object is the path it traces out on a spacetime diagram — the complete record of where it was at every moment. A point on the diagram is an *event*.

With ct on the vertical axis, a light signal moves one unit of distance per unit of ct, so **light travels along the** 45° **lines** $x = \pm ct$. An object at rest stays at fixed x, so its worldline is *vertical*. An object moving at speed u has a worldline of slope c/u, and since no object moves faster than light, $u < c$ means that **every worldline is steeper than** 45°. The faster an object moves, the more its worldline tilts toward the 45° light line — but it can never reach it.

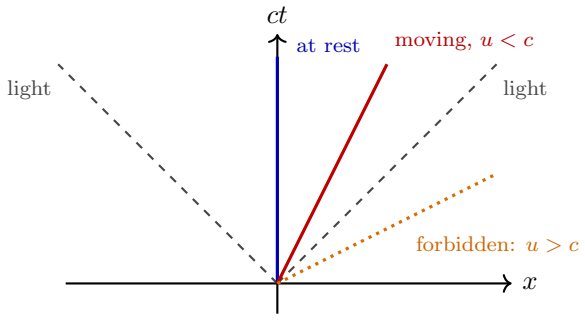


Figure 1: A spacetime diagram. Light moves along the 45° dashed lines. An object at rest has a vertical worldline; a moving object's worldline tilts, but always remains steeper than 45°. A worldline shallower than 45° would mean $u > c$ and is forbidden.

4 Order of Events and Causality

Can the temporal order of two events be reversed by changing reference frame? Suppose $\Delta t > 0$ in frame O (event 2 occurs after event 1, according to O). The order is reversed in O' when $\Delta t' < 0$:

$$\Delta t' = \gamma\Big(\Delta t - \frac{v}{c^2}\,\Delta x\Big) \implies \Delta t' < 0 \quad \text{when} \quad \frac{v}{c^2}\,\Delta x > \Delta t, \tag{9}$$

that is, when

$$\frac{\Delta x}{\Delta t} > \Big(\frac{c}{v}\Big)\,c. \tag{10}$$

According to special relativity, effects can propagate at most at the speed of light, $\leq c$. Hence if event 2 is *caused* by event 1,

$$\frac{\Delta x}{\Delta t} \leq c. \tag{11}$$

Also, observers can move only at speeds $v \leq c$, so $c/v \geq 1$. Therefore the reversal condition

$$\frac{\Delta x}{\Delta t} > \Big(\frac{c}{v}\Big)\,c \;\geq\; c \tag{12}$$

cannot be satisfied when event 2 is caused by event 1.

Conclusion. The order of two events can be switched by a sufficiently fast observer, but only when neither event causes the other. Causally connected events keep their order for all observers. For example, all observers agree that the spaceship took off before it landed, and that the gun was fired before the target was struck.

The light cone

These statements become a picture. The two light lines through an event E carve spacetime into three physically distinct regions. (Restoring the second spatial dimension turns the 45° lines into a cone, which is where the name comes from.)

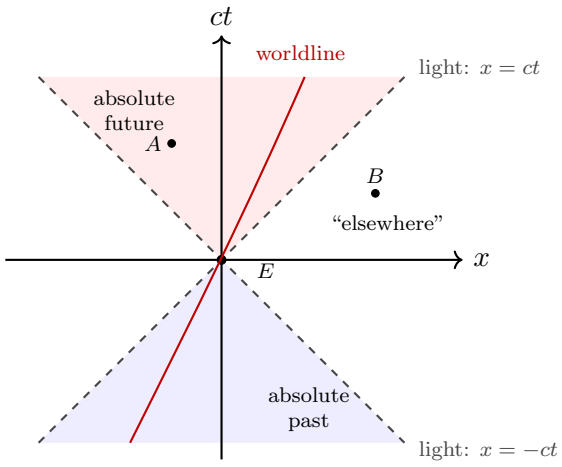


Figure 2: The light cone. The 45° light lines through an event E divide spacetime into the absolute future of E, the absolute past of E, and "elsewhere." Event A lies inside the future cone: it can be caused by E, and every observer agrees it happens after E. Event B lies outside the cone: no signal can connect it to E, and its time order relative to E depends on the observer.

- **Absolute future** ($|\Delta x| \le c\,\Delta t$, $\Delta t > 0$): events reachable from E by a signal travelling at $\le c$. They can be *caused* by E, and by the result above, *all* observers agree they occur after E. These are the *effects* of E.
- **Absolute past** ($|\Delta x| \le c\,|\Delta t|$, $\Delta t < 0$): events that could have caused E. All observers agree they occur before E. These are the possible *causes* of E.
- **"Absolute elsewhere"** ($|\Delta x| > c\,|\Delta t|$): no signal can connect these events to E. This is precisely the region $|\Delta x|/|\Delta t| > c$ where the time order is frame-dependent — harmlessly so, since neither event can influence the other.

Notice that on the slice $t = 0$ it is the two *cones* that pinch down to the single point E: every other event at that instant lies in the elsewhere region. That is the geometric statement that nothing at a distance can influence E instantaneously.